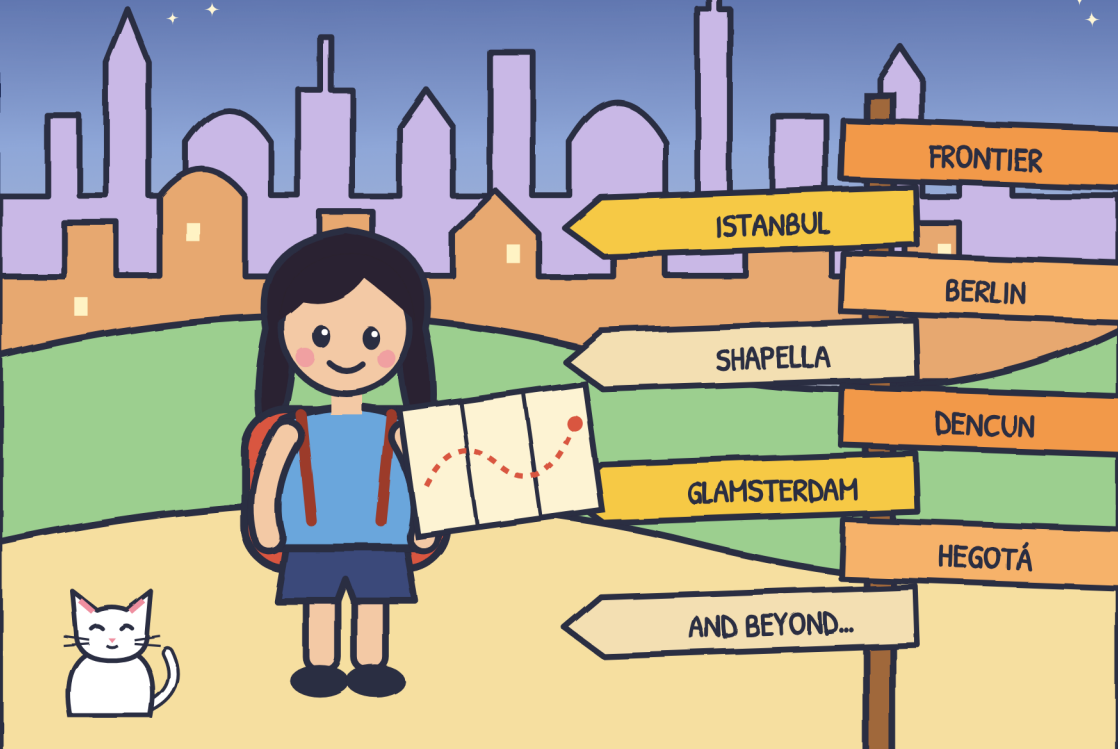
Cities, Stars & Dragons

...and other things
Ethereum named its upgrades







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Preface

This book tells the stories of how Ethereum upgrade names came to be.

It is based on public information from Ethereum Improvement Proposals (EIPs), core developer meetings, official announcements and community discussions, along with my own experiences and memories from years of helping coordinate Ethereum upgrades.

It is written in a simple and friendly way, with illustrations, so that people of all ages can enjoy it. This book is for anyone who is curious about Ethereum, whether you are new to the ecosystem, a long-time contributor, or someone who just likes a good story.

Upgrade names are more than just labels. They are part of Ethereum's journey and reflect the people, ideas, and conversations that shape it.

I hope this book inspires you to imagine what comes next, and to be part of Ethereum's future.



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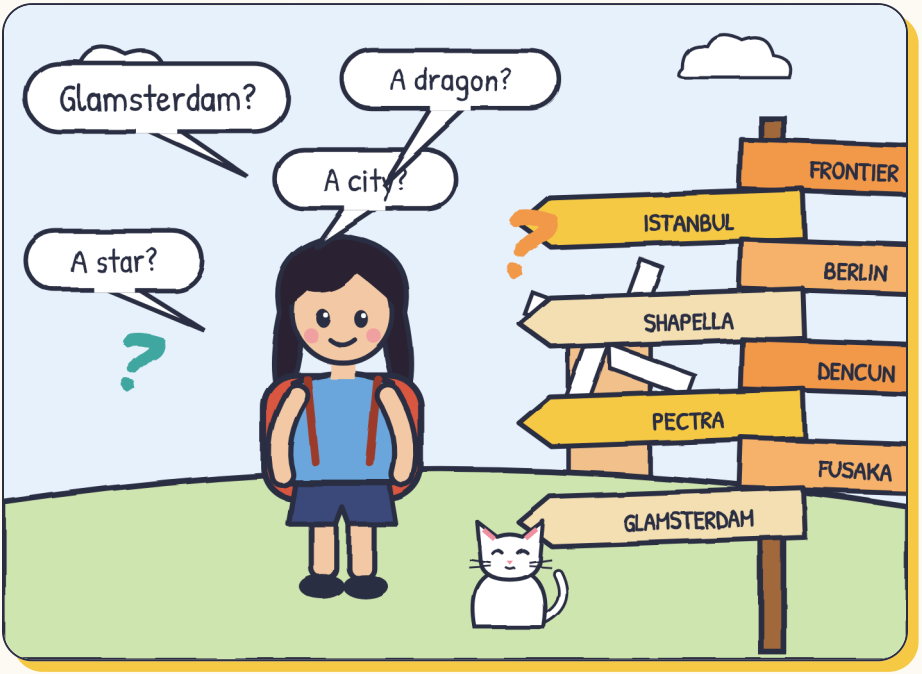
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1

What on Earth Is Glamsterdam?



Hello, Etherean! ✦

Our journey begins with a rather unusual name: Glamsterdam. Just a little curiosity is all you need.

If you're new to Ethereum or have been following its development from a distance, your first question might be: What on earth is **Glamsterdam**? Is it a city, a music festival, or perhaps a new destination on the map?

Well, it is the name of an upcoming Ethereum network upgrade. Glamsterdam is not a real city but a combination of two names: Gloas, a star representing the Consensus Layer, and Amsterdam, the city that hosted Devconnect in 2022, representing the Execution Layer.

But why does an Ethereum upgrade need two names? Why are some upgrades named after cities, others after stars, and some a combination of both? And who decides what Ethereum's next upgrade should be called?

These are some of the questions that led me to explore the history behind Ethereum's upgrade names.

For someone new to Ethereum, it helps to know that Ethereum is a decentralized blockchain network supported by developers, researchers, validators, and communities around the world. Like other software, it needs regular upgrades to improve security, performance, scalability, and user experience. But changing Ethereum is not as simple as releasing a new version of an application. It involves research, proposals, discussions, testing, and coordination among independent teams.

In a PEEPanEIP episode, Ben Edgington, an Ethereum researcher and client developer, compared Ethereum's development to a bazaar rather than a cathedral. Unlike a carefully planned structure built according to a fixed design, Ethereum evolves through an open process, with people contributing ideas, challenging assumptions, and adapting plans as they learn.

As Ben put it:

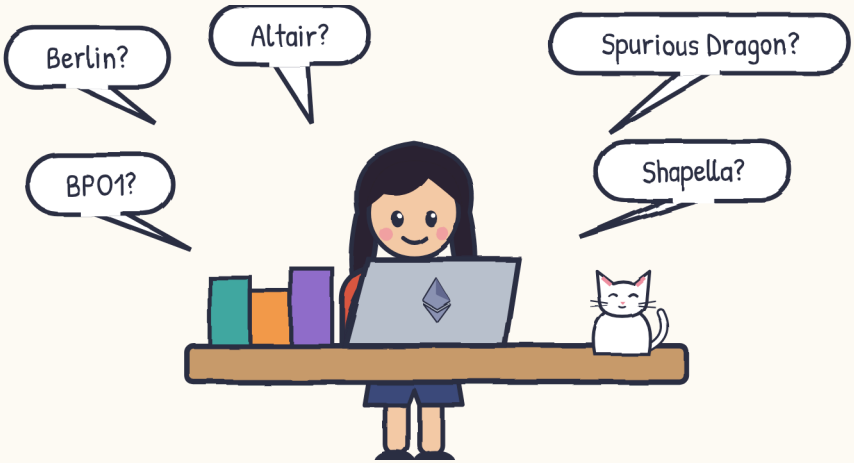
> "We're always moving, always aspiring, always got better places to go."

That is part of what makes Ethereum interesting. Its roadmap has changed many times over the years, but the community continues to work towards improving the network.

Ethereum Improvement Proposals, or EIPs, help document these changes. The process of considering proposals for network upgrades is described in **EIP-7723**. And because an upgrade can include several EIPs affecting different parts of Ethereum, having a recognizable name helps everyone follow its progress.

As of October 2026, Glamsterdam has activated on Sepolia, but the mainnet activation date has not yet been confirmed. The upgrade is focused on changes to block production, gas pricing and Ethereum's capacity.

To understand how we arrived at a name like Glamsterdam, we'll travel back to Ethereum's early days. Before the cities, before the stars, and before developers started combining the two. By the end of this journey, I hope these unusual names will make a little more sense, along with the people, discussions, and decisions behind them.

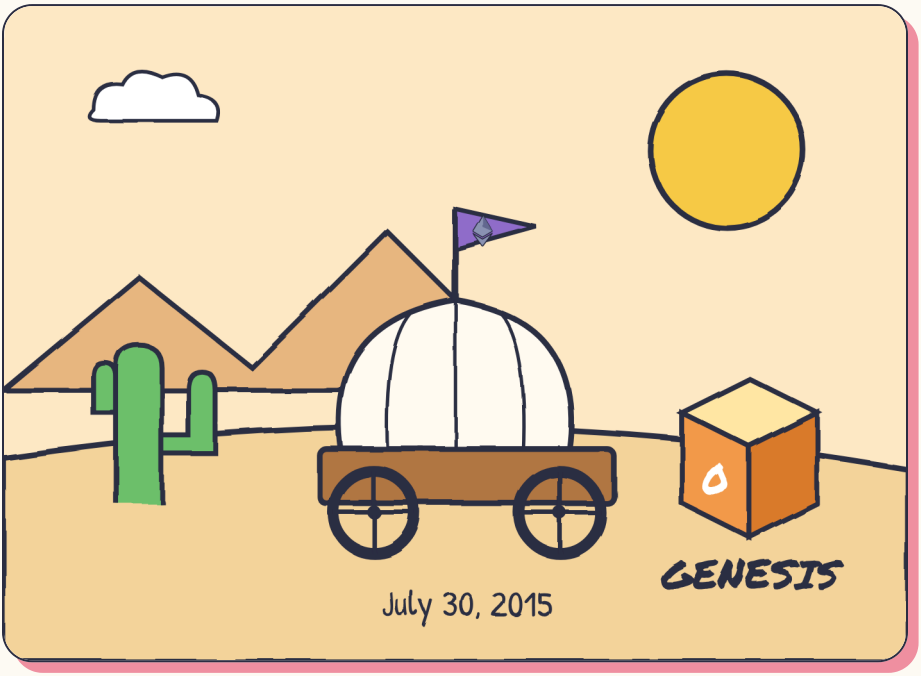


FUN FACT ✦

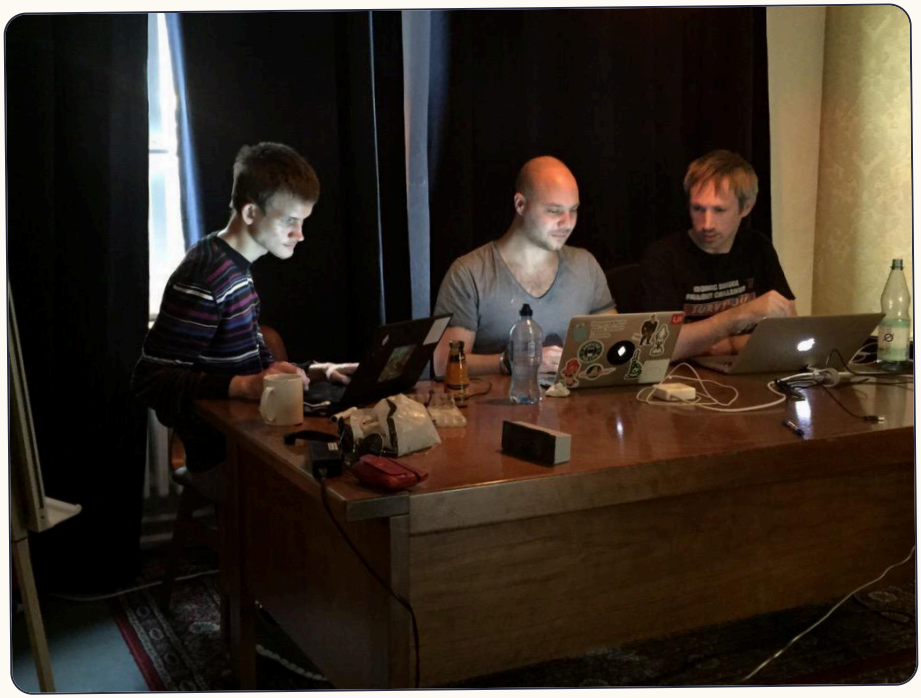
Glamsterdam is a portmanteau, a word formed by combining parts of two other words. Ethereum has several such upgrade names, including Shapella, Dencun, Pectra, and Fusaka. We'll discover how each got its name along the way.

2

Welcome to the Frontier



Before the cities, before the stars: a wagon, a genesis block and a brave first step.



Early days of Ethereum development. Image: Ethereum Foundation Blog, "Homestead Release," February 29, 2016, blog.ethereum.org/2016/02/29/homestead-release

Ethereum's development began in December 2013, when two developers and a college dropout researcher came together to work on an ambitious project idea. Soon, more like-minded people joined them, helping turn that idea into building a programmable blockchain. The first Proof of Concept (PoC) was released on February 1, 2014. By the time the Ether crowdsale began in July, the team had reached PoC 5, with multiple software clients already working towards compatibility.

In the February 2016 **Homestead** announcement, Jeffrey Wilcke recalled the excitement of those early days. The team was thrilled. Building Ethereum was one thing, but seeing people believe in the idea and support its development was something else. Development continued through 2014 and 2015, with Jeffrey Wilcke, Gavin Wood, Vitalik Buterin, and many other contributors working towards the first public release.

On July 30, 2015, Ethereum launched its first live network, called Frontier.

✦ Why Frontier?

The name **Frontier** was inspired by the American Frontier, a period when settlers explored new territories, established communities, and faced many unknown challenges.

Ethereum's early developers saw a similar journey ahead. In the July 2015 launch announcement, Stephan Tual compared early adopters to settlers of the American Frontier. Both had opportunities to explore something new, but they also had to be prepared for the risks.

> "Like their counterparts during the American Frontier, these settlers will be presented with vast opportunities, but will also face many dangers."

The name suited this stage of Ethereum's development. Frontier was a working blockchain, but the software was still basic. The release was mainly intended for developers, miners, and technically experienced users. Most interactions required command-line tools, and the wider community was encouraged to wait for a more user-friendly release.

The first block, known as the Genesis block, marked the beginning of Ethereum's blockchain. From there, the network began producing blocks and processing transactions.

✦ Frontier Thawing

Ethereum's early developers were careful about opening the network too quickly. Frontier initially had a hardcoded gas limit of just 5,000 gas per block, which was too low for ordinary transactions. This gave miners and node operators time to get their software running before the network became fully active.

The next step was called **Frontier Thawing**, activated in September 2015. It removed the initial gas restriction, allowing more transactions and smart

contract activity. The name followed naturally from Frontier. Ethereum had launched, but some restrictions were still in place. With Thawing, the network gradually became more usable.

✦ From Frontier to Homestead

Ethereum's early roadmap described four development stages: Frontier, Homestead, **Metropolis**, and Serenity. Each represented a step towards a more mature network, with improvements to functionality, security, and user experience.

The next milestone was Homestead, activated on March 14, 2016, at block 1,150,000. The name continued the Frontier theme. Just as settlers move from exploring new territory to establishing homes, Ethereum was moving beyond its experimental stage towards a more stable network.

In the original launch plan, Homestead was expected to follow once developers and security auditors were satisfied with Frontier's stability. By October 2015, Vitalik Buterin was discussing Ethereum's progress towards a more mature platform, with improvements to security, scalability, and the protocol. In his February 2016 announcement, Jeffrey Wilcke recalled how the community had worked together to resolve two consensus bugs after Frontier launched. The team was now confident enough to remove the earlier safety warning from Ethereum's website and move into Homestead.

Homestead introduced improvements through EIP-2, EIP-7, and EIP-8, covering transaction validation, smart contract functionality, and communication between Ethereum clients.

The Ethereum Foundation later described this period as Ethereum growing from a small homestead into a bustling metropolis. More people were joining, applications were becoming more complex, and the network was finding new uses. The names reflected that journey, from exploring a new frontier to building an ecosystem that was growing beyond its early beginnings.

Years later, the cities hosting these gatherings became part of Ethereum's upgrade naming tradition. We'll return to that story when Ethereum gets its passport in Chapter 5.

THE RECORD ✦ Ethereum's First Upgrades

Upgrade	Mainnet activation	Block	What it did
Frontier	July 30, 2015	Genesis	Initial network launch
Frontier Thawing	Sept 7, 2015	200,000	Initial gas-limit restriction lifted
Homestead	March 14, 2016	1,150,000	EIP-2, EIP-7, EIP-8

This was just the beginning of the new ecosystem. It meant people were ready to build more of it.

A DISCOVERY FROM THE EARLY DAYS ✦

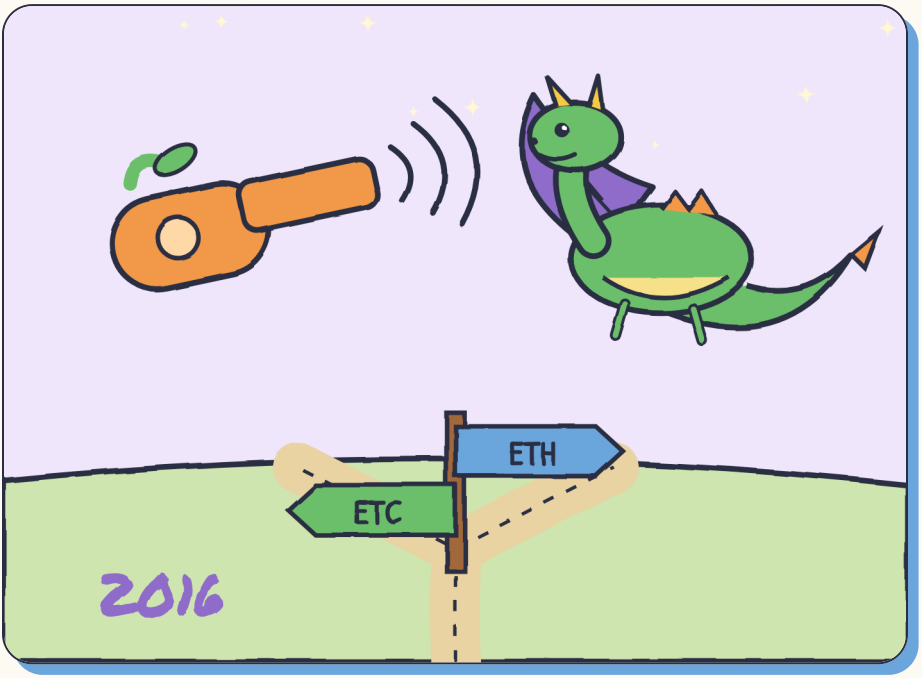
The first Frontier release temporarily limited blocks to 5,000 gas. This gave miners time to begin and early users time to set up their software. Even a new frontier needs a careful first step.

FUN FACT ✦

Homestead activated on March 14, also known as Pi Day. A fitting date for a network built on mathematics!

3

A Whistle, a Dragon and a Difficult Choice



In 2016, Ethereum faced some difficult decisions. And somewhere along the way, a whistle and a dragon joined its upgrade history.

Ethereum was still a young network when The DAO, an early experiment in decentralized decision-making, attracted considerable attention. Created by the team behind Slock.it, The DAO was designed to let people pool funds and decide which projects to support through smart contracts rather than a traditional organization.

Between April and May 2016, The DAO raised approximately 12 million ETH, worth around $150 million at the time. It was an extraordinary amount for such a young ecosystem. But before The DAO could begin funding projects, something went wrong.

On June 17, 2016, an attacker exploited a vulnerability in its smart contracts and diverted approximately 3.6 million ETH, nearly one-third of its funds.

This led to one of Ethereum's earliest and most difficult disagreements. Should the network introduce a change to recover the funds, or should the transactions remain as they were? Some supported a hard fork to recover the funds, while others believed that changing the blockchain's history went against the principle of immutability. A soft fork was also considered, but security concerns made that option unsuitable.

The Ethereum Foundation invited the wider community to participate in the decision. In a July 2016 blog post titled To Fork or Not to Fork, Jeffrey Wilcke explained the options and how the community could express its preference. After considerable discussion, the **DAO Fork** activated on July 20, 2016, at block 1,920,000. However, not everyone supported the change. Those who preferred to continue with the original chain formed the community around Ethereum Classic (ETC), while the forked chain continued as Ethereum (ETH).

The name DAO Fork came directly from the event that led to the upgrade. It was also an early example of how disagreements over protocol changes could lead to different groups choosing different paths.



FROM THE CODE CONFESSION ✦

Remembering the DAO Fork

I had a memorable conversation with Thomas Jay Rush on my series The Code Confession. Thomas had been involved with Ethereum since late 2015 and remembered the excitement around smart contracts and The DAO.

He recalled how Alexey (Erigon) posted a message during the attack:

> "I think the DAO is being drained."

Alexey then took a train to work and, according to Thomas, returned nearly 45 minutes later to confirm his concerns. Imagine watching something like this unfold through messages while the community was trying to understand what was happening!

Thomas also remembered waking up after the DAO Fork to find his ETH holdings recorded on two separate chains.

> "I have the same number of tokens on one chain, the same number of tokens on another chain."

His experience gives us a glimpse of what the fork meant to people who were involved at the time. One network had become two, and both continued with their own communities. The idea that one blockchain could split into two networks, each with its own tokens and community, was quite an experience for someone who had joined Ethereum only months earlier.

✦ One More Chapter in The DAO Story

The DAO incident also attracted regulatory attention. In July 2017, the U.S. Securities and Exchange Commission (SEC) published an investigation report concluding that DAO Tokens were securities under U.S. federal securities laws. Although the SEC did not pursue an enforcement action in this particular case, the report became an important reference for blockchain projects.

The DAO had started as an experiment in organizing people and funds through code. Its story raised questions about security, governance, community decisions, and regulation that continued well beyond the 2016 fork.

✦ And Then Came Tangerine Whistle

Later in 2016, Ethereum faced another problem. A series of denial-of-service (DoS) attacks slowed the network. Some operations required more computing resources than their gas costs reflected, making it relatively inexpensive for attackers to overload Ethereum nodes.

Developers responded with an emergency upgrade called **Tangerine Whistle**, activated on October 18, 2016, at block 2,463,000. Through EIP-150, it increased the gas costs of certain operations to make these attacks more expensive.

But where did the name Tangerine Whistle come from?

The original Meta EIP, EIP-608, also calls the upgrade Anti-DoS, which describes its purpose. The name Tangerine Whistle is recorded in Ethereum's history, but I haven't found a historical explanation of who chose it or why.

One could imagine the whistle being an emergency alarm, but that would be my interpretation rather than a documented reason. And the Tangerine part remains a mystery.

✦ The Spurious Dragon

Tangerine Whistle helped address the immediate attacks, but there was still some cleanup to do. The attacks had left behind many empty accounts, adding unnecessary data to Ethereum's state.

Another upgrade followed on November 22, 2016, called **Spurious Dragon**.

It included changes to remove empty accounts, adjust gas costs, and improve transaction security. EIP-161 addressed empty accounts, while EIP-155 introduced replay protection, an important improvement following the DAO-related chain split.

The original Meta EIP, EIP-607, also records State-clearing as an alternative name. This describes the cleanup work involved in the upgrade.

The word spurious means false or not genuine, which seems to fit the unwanted accounts left behind by the attacks. But I haven't found an original explanation confirming that this was why the name was chosen. And why a dragon? That part remains unclear too.

By December 2016, Ethereum had completed these two upgrades and the subsequent state cleanup. Development continued towards its next major phase, **Metropolis**.

In short, the DAO Fork involved a difficult community decision, while Tangerine Whistle and Spurious Dragon were responses to attacks on the network.

We know why these upgrades were needed, even if we don't yet know the complete stories behind their names. Perhaps someone who was involved in those early discussions still remembers!

THE RECORD ✦ ETHEREUM'S 2016 UPGRADES

Upgrade	Date	Block	Related EIPs
DAO Fork	July 20, 2016	1,920,000	EIP-779
Tangerine Whistle	Oct 18, 2016	2,463,000	EIP-150 (Meta: EIP-608)
Spurious Dragon	Nov 22, 2016	2,675,000	EIP-155, 160, 161, 170 (Meta: EIP-607)

ETHEREAN DISCOVERY ✦

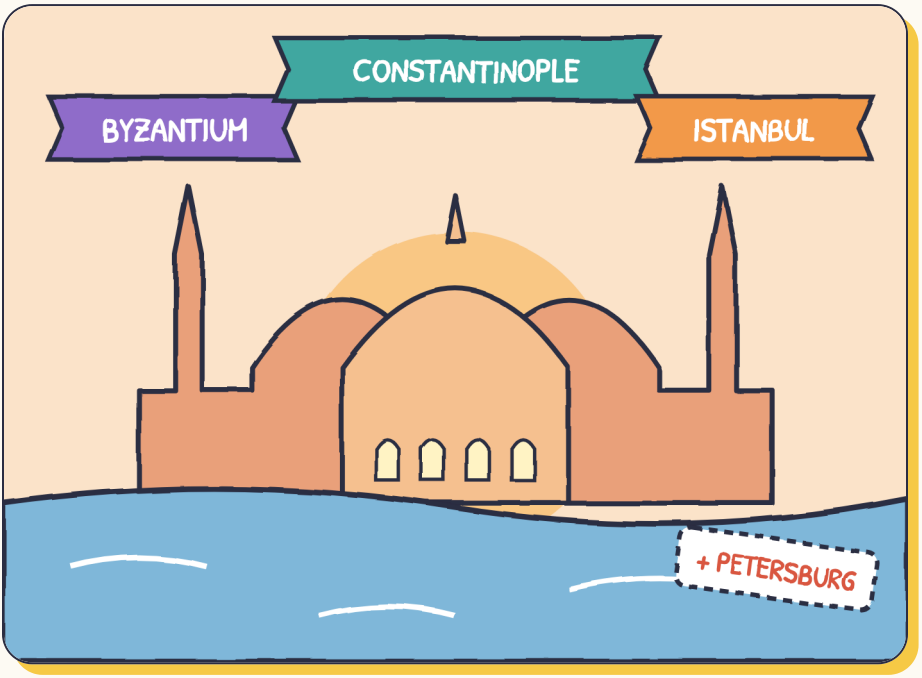
Not every hard fork leaves two lasting communities. The DAO fork did, giving us Ethereum and Ethereum Classic. The name "DAO fork" was direct: it described the event rather than a mythical creature.

FUN FACT ✦

Tangerine Whistle and Spurious Dragon were also known as Anti-DoS and State-clearing. Those names explain their technical purpose, even though the origins of the more memorable names remain uncertain.

4

A Tale of Three Cities (and One Extra)



Our next stop takes us through three names from the history of one city, with an unexpected fourth along the way.

Byzantium. Constantinople. Istanbul.

These three names are connected to the history of the same city, known by different names over centuries. Ethereum borrowed them for network upgrades during a major development phase called **Metropolis**.

Byzantium, activated on October 16, 2017, was the first part of Metropolis. It introduced improvements to Ethereum's cryptography and virtual machine, reduced mining rewards, and delayed the **difficulty bomb**.

Next came **Constantinople**, planned as the second part of Metropolis. It included improvements to smart contract functionality, lower gas costs for certain operations, another reduction in mining rewards, and a further delay to the difficulty bomb.

But Constantinople had an unexpected turn.

✦ The Upgrade That Almost Happened

Constantinople was scheduled for January 2019. Just 37 hours before activation, security researchers at ChainSecurity reported a vulnerability involving EIP-1283. The proposed change could make certain smart contracts vulnerable to reentrancy attacks, potentially allowing funds to be withdrawn in ways their developers had not intended.

With so little time left, developers had to make a quick decision. Discussions started across Gitter, Telegram, and an emergency call with key stakeholders. Within a few hours, the decision was made to postpone the upgrade.

I joined Ethereum Cat Herders around the time of these upgrade discussions, so the coordination and documentation from this period are close to my heart. The team later published a post-mortem report documenting what happened, who was involved, and what could be improved.

What stayed with me was how quickly people came together to address the problem. The upgrade was postponed, the issue was investigated, and a revised plan was prepared.

✦ And Then Came Petersburg

When Constantinople finally activated on February 28, 2019, it arrived alongside another upgrade called **Petersburg**. Both activated at the same block, 7,280,000.

Why two upgrades at once?

Some Ethereum testnets had already implemented the original Constantinople changes, including EIP-1283. To address the security concern while keeping the network rules consistent, developers introduced Petersburg to disable EIP-1283. This allowed the remaining Constantinople improvements to go ahead without introducing the vulnerable change on mainnet.

It was a marvellous piece of dev work: the upgrade stayed on schedule, and the simultaneous upgrade removed the problematic EIP, keeping the network safe.

To me, this is a great example of Ethereum's coordination. Developers managed to preserve the planned improvements while addressing a last-minute security issue. Two upgrades at the same block, with one removing a change introduced by the other!

Later that year, **Istanbul** activated on December 7, 2019, continuing the historical city theme with further improvements to Ethereum's security, performance, and gas costs.

So, Byzantium, Constantinople, and Istanbul shared a historical connection, while Petersburg became the unexpected extra name because of a technical issue.

These names came before Ethereum adopted its more familiar Devcon city-naming tradition. That story begins in our next chapter.

THE RECORD ✦ Early City Names

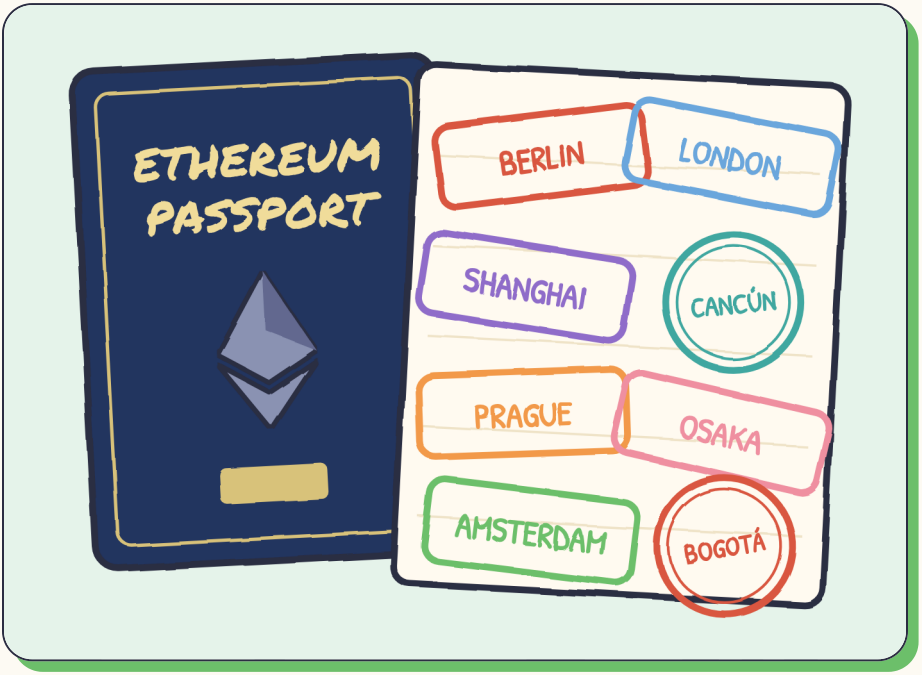
Upgrade	Mainnet activation	Meta EIP
Byzantium	October 16, 2017	EIP-609
Constantinople	February 28, 2019	EIP-1013
Petersburg	February 28, 2019	EIP-1716
Istanbul	December 7, 2019	EIP-1679

FUN FACT ✦

Byzantium, Constantinople, and Istanbul are names associated with the same historic city, now Istanbul in Türkiye. But Petersburg is a different city altogether, in Russia. Ethereum's naming history had taken an unexpected detour!

5

Ethereum Gets a Passport



Why should software carry city names? The story begins with a meeting.

In August 2019, Ethereum developers were discussing what to call the next network upgrade. Until then, upgrades had different kinds of names, but there was no consistent naming tradition. With more upgrades being planned, having a recognizable naming pattern could make things easier for everyone.

During ACD#68, Alex Beregszaszi suggested naming future upgrades after cities that had hosted Devcon, Ethereum's developer conference. Instead of calling an upcoming upgrade **Istanbul** 2, why not call it **Berlin**, the city that hosted Devcon 0? After some discussion, the group agreed to use Devcon city names for public-facing upgrades.

I love this part of Ethereum's history. Even something as simple as naming an upgrade was discussed openly, with different opinions recorded in the meeting notes.

✦ Berlin, the First City Stamp

Berlin became the first upgrade under this naming convention, activating on April 15, 2021. The name came from Devcon 0, held in Berlin in 2014.

Berlin included four EIPs. EIP-2565 adjusted the gas cost of certain mathematical operations, while EIP-2929 increased gas costs for some operations that access Ethereum's stored data. EIP-2930 introduced optional access lists, helping transactions manage these changes. And EIP-2718 introduced a new way to support different transaction types, making future improvements easier.

I remember discussing two of these proposals, EIP-2929 and EIP-2930, with Vitalik Buterin and Martin Holst Swende in a PEEPanEIP episode. The idea was to better account for the work Ethereum nodes perform when accessing stored information, while reducing the risk of certain attacks that could slow down the network.

On improving Ethereum's security, Vitalik said,

> "EIP-2929... basically just um solving the kind of last remaining uh dos vulnerability that we know about in the Ethereum chain."

I also discussed EIP-2718 with its author, Micah Zoltu. What stood out to me was that this proposal was not about adding a feature users would immediately notice. It was about making future changes easier to introduce without adding unnecessary complexity to Ethereum.

Micah shared a useful piece of advice for EIP authors:

> "My number one recommendation for EIP authors is keep your EIP as simple as possible."

Sometimes, improvements that users barely notice can make a big difference to how Ethereum evolves.

A LITTLE BERLIN MEMORY ✦

I also remember helping with Ethereum Cat Herders' early upgrade countdown livestreams with Hudson Jameson. During one of them, I missed the moment to start the stream and had to repeat the introduction!

Not everything in Ethereum runs perfectly on the first try, including our livestreams. :)

✦ A Passport Full of Stamps

After Berlin came **London**, named after the city that hosted Devcon 1. Later upgrades continued the tradition with Shanghai, Cancun, Prague, and Osaka, all cities that had hosted Devcon.

The convention eventually expanded to include Devconnect locations, bringing Amsterdam into Ethereum's naming history. Bogotá, which hosted Devcon 6, also joined the list of upgrade names.

There was one interesting exception. During **The Merge**, the Execution Layer transition was called **Paris**, a reference to EthCC, the Ethereum Community Conference.

Of course, Ethereum wasn't actually traveling between these cities. The names were a way to connect upgrades with places where developers, researchers, and community members had gathered over the years.

I like this naming tradition because it reminds us that Ethereum may be a global network, but the people working on it also meet in real places, exchange ideas, disagree, and build things together.

THE RECORD ✦ ETHEREUM'S CITY NAMES

City	Ethereum event	Upgrade
Berlin	Devcon 0, 2014	Berlin
London	Devcon 1, 2015	London
Shanghai	Devcon 2, 2016	Shapella
Cancun	Devcon 3, 2017	Dencun
Prague	Devcon IV, 2018	Pectra
Osaka	Devcon V, 2019	Fusaka
Amsterdam	Devconnect, 2022	Glamsterdam
Bogotá	Devcon 6, 2022	Hegotá
Paris	EthCC (the exception)	The Merge

Note: In combined upgrades such as Shapella and Dencun, the city names represent the Execution Layer. The other half comes from a star name, which we'll explore in later chapters.

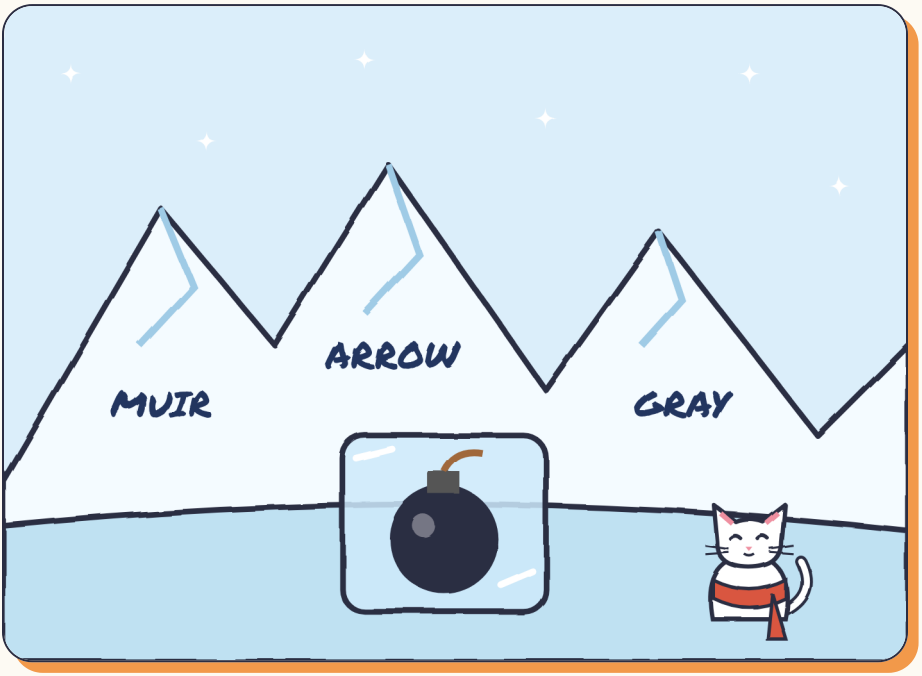
And that is how Ethereum started collecting passport stamps without ever leaving the internet!

FUN FACT FROM THE TRAIN ✦

Back in 2014, Jeffrey Wilcke and Gavin Wood first discussed the idea of an Ethereum developers' conference while travelling by train to Zurich Airport. Years later, the cities hosting these gatherings became part of Ethereum's upgrade naming tradition.

6

The Great Ethereum Ice Age



Our next stop is the Ethereum Ice Age. Don't worry, no winter clothes needed!

Before **The Merge**, Ethereum used Proof of Work, where miners competed to produce blocks. The protocol also had something called the **difficulty bomb**, a mechanism designed to gradually make mining more difficult until block production became extremely slow. This period of increasing block times was known as the Ethereum **Ice Age**.

The idea behind the difficulty bomb was to encourage Ethereum's planned transition from Proof of Work to Proof of Stake. But developing and testing the new system was taking longer than expected. If the bomb was allowed to progress, Ethereum would slow down before the new system was ready. Developers needed more time, which meant delaying the bomb.

✦ And Then Came the Glaciers

Ethereum had already started naming major upgrades after Devcon cities. But for upgrades focused only on delaying the Ice Age, a different naming tradition developed. The developer community began using names of real glaciers, particularly those known for retreating, to reflect the purpose of these upgrades.

The first was **Muir Glacier**, activated in January 2020. Through EIP-2384, it delayed the difficulty bomb, giving developers more time to prepare for Proof of Stake. It was named after Muir Glacier in Alaska, known for its significant retreat over the years.

Next came **Arrow Glacier** in December 2021, delaying the bomb again through EIP-4345. Like Muir Glacier, it was a focused upgrade with one main purpose: pushing back the Ice Age until The Merge arrives.

Gray Glacier followed in June 2022 through EIP-5133, just a few months before The Merge. Its name had a clever connection. The real Gray Glacier merges into another glacier, just as Ethereum was preparing to merge its Execution Layer with the **Beacon Chain**. The Ethereum Foundation even mentioned this connection in its upgrade announcement.

Three glaciers, three delays, all giving Ethereum more time to prepare for one of its biggest changes.

FROM PEEPANEIP ✦

Why Have a Difficulty Bomb?

I discussed the difficulty bomb in two PEEPanEIP episodes, one with James Hancock about EIP-3554 and another with T. Jay Rush about EIP-4345. Both conversations helped explain why Ethereum had introduced this mechanism in the first place.

James explained that the difficulty bomb was also a way to keep Ethereum's upgrade schedule moving. Without a reason to act, difficult decisions could keep getting postponed. As he put it:

> "Having a we need to do something by this date means we have a very easy way to say any EIP that's ready by this time can make it in."

T. Jay Rush offered another perspective. He saw the difficulty bomb as a way for the community to hold core developers accountable, because someone had to update the software before the network slowed down.

> "It gives the community a forcing function against the core devs."

These conversations showed me that the difficulty bomb was not just a technical feature. It also played a role in Ethereum's coordination and governance, encouraging people to revisit decisions and keep working towards the transition.

Of course, delaying the bomb was never meant to be a permanent solution. In September 2022, The Merge finally happened. Ethereum moved to Proof of Stake, and the old Proof-of-Work difficulty bomb was no longer needed.

THE RECORD ✦ ETHEREUM'S GLACIERS

Upgrade	Date	EIP	Naming connection
Muir Glacier	January 2020	EIP-2384	Part of Ethereum's Ice Age naming tradition
Arrow Glacier	December 2021	EIP-4345	Another delay to the difficulty bomb
Gray Glacier	June 2022	EIP-5133	Named after a glacier that merges with another, referencing The Merge

The Ethereum Foundation explained the reason for Gray Glacier's name: the real Gray Glacier merges into another glacier, a fitting reference to Ethereum's approaching Merge. No real ice formed inside the blockchain. The glaciers were names for upgrades that bought time. The bomb was software logic. For once, a naming story comes with a documented explanation!

A DISCOVERY ABOUT THE GLACIERS ✦

These upgrades mainly gave developers more time before the difficulty bomb slowed mining. Gray Glacier arrived just a few months before The Merge made the old Proof-of-Work difficulty bomb unnecessary.

FUN FACT ✦

Before The Merge, Ethereum upgrades were scheduled by block number, making the exact activation time difficult to predict. Block times could vary, so even when developers announced an expected date, the network followed its own schedule.

7

When Ethereum Started Counting Stars



Look up! Ethereum's next naming adventure takes us from Earth to the stars.

While Ethereum's original chain was still running on Proof of Work, another part of its future was taking shape. The **Beacon Chain** launched on December 1, 2020, laying the foundation for Ethereum's move to Proof of Stake.

When it was time for the Beacon Chain's first upgrade, the community needed a new naming approach. After some discussions and even emoji voting, client teams decided to name Consensus Layer upgrades after stars, following alphabetical order. A simple idea that gave Ethereum a whole new universe of names to explore!

The first was **Altair**, which activated on October 27, 2021. Altair is the twelfth brightest star in the night sky, located in the constellation Aquila. It also rotates remarkably fast, at around 286 kilometres per second. Quite an energetic choice for Ethereum's first Consensus Layer upgrade.

But Altair was more than a lovely star name. It was the first time the live Beacon Chain underwent a network upgrade. It introduced improvements to validator rewards and penalties, along with support for light clients, which help devices like mobile phones follow Ethereum without downloading the entire blockchain.

FROM PEEPANEIP ✦

I remember discussing Altair in multiple PEEPanEIP episodes, one with Vitalik Buterin, Danny Ryan, Alex Stokes and other developers. These conversations helped explain why the first Beacon Chain upgrade mattered beyond its name.

Danny described Altair as a "warm-up fork" before The Merge. Developers wanted to gain experience upgrading the live Beacon Chain before attempting the much bigger transition that would bring Ethereum's two layers together.

Vitalik also discussed light clients and how they could follow Ethereum with very little data. At one point, on the definition and risk of discouragement attacks, he explained:

> "The idea of a discouragement attack is basically like to use a griefing attack so an attack that causes another validator to either lose rewards or suffer a penalty with the goal of... increasing the chance that that validator drops out."

In another conversation on accounting reforms, Alex Stokes walked us through changes to how validators were rewarded for participating in the network. These improvements made the system more efficient and easier for developers to maintain.

> "...the idea of the reforms was more to maintain behavior but keep it either improve the efficiency like we talked about or again make it easier to reason about either from a spec or an implementation standpoint."

So, while the name came from the stars, Altair was doing some important work here on Earth!

✦ And Then Came Bellatrix

With Altair taking the letter A, the next upgrade needed a star beginning with B.

A public GitHub discussion brought together suggestions for the next name. People proposed candidates, discussed the choices, and eventually settled on **Bellatrix**, a star in the constellation Orion.

Bellatrix would later become an important part of Ethereum's history for **The Merge**. But we'll come to that story in the next chapter.

The alphabetical tradition continued with **Capella**, Deneb, Electra, Fulu, Gloas, and **Heze**. Each name followed the same idea, making the sequence easy to recognize and keeping it separate from the city names used for Execution Layer upgrades.

For a while, Ethereum had two naming maps. One took us through cities that hosted Devcon, while the other took us across the night sky.

And eventually, these two worlds would meet.

THE RECORD ✦ ETHEREUM'S STAR NAMES

	Star	Constellation	Ethereum connection
A	Altair	Aquila	First Beacon Chain upgrade, 2021
B	Bellatrix	Orion	Consensus Layer preparation for The Merge
C	Capella	Auriga	Paired with Shanghai to form Shapella
D	Deneb	Cygnus	Paired with Cancun to form Dencun
E	Electra	Taurus	Paired with Prague to form Pectra
F	Fulu	Cassiopeia	Paired with Osaka to form Fusaka
G	Gloas	Pisces	Consensus Layer name for Glamsterdam
H	Heze	Virgo	Consensus Layer name for Hegotá

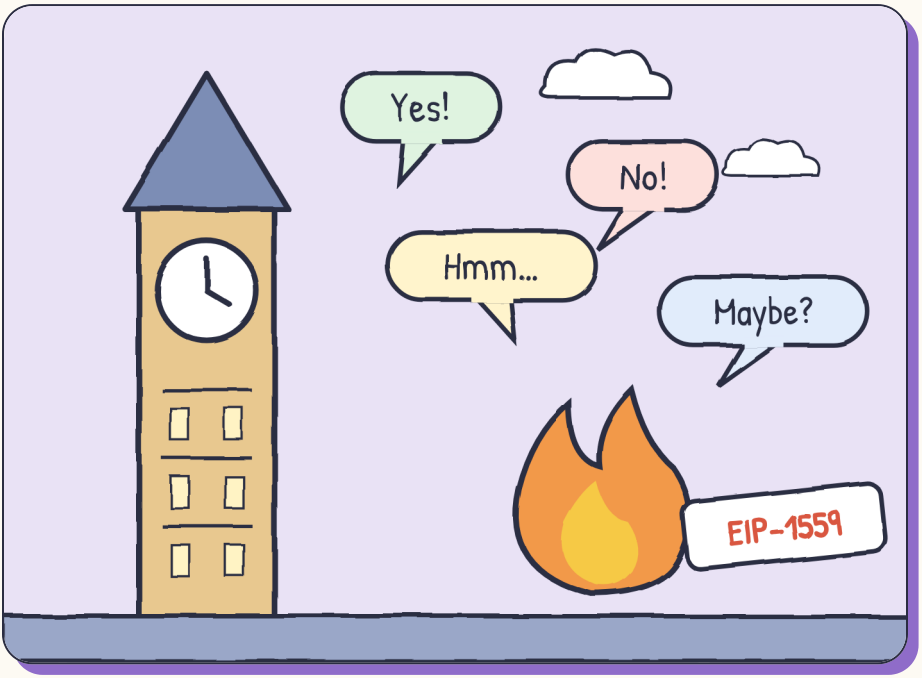
The final two names refer to upgrades under development as of October 2026. The naming sequence is documented in EIP-8133.

A TINY GALAXY OF NAMES ✦

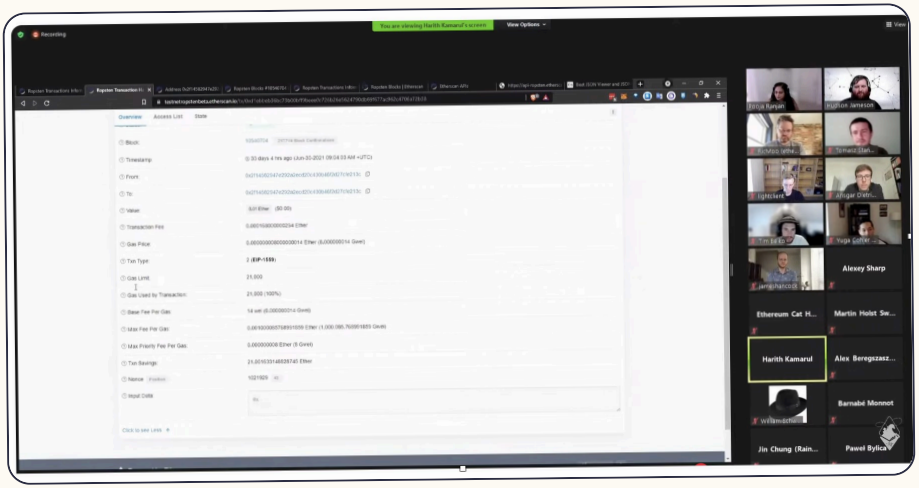
Our own Milky Way has far more stars than Ethereum will ever need for upgrade names. But an alphabet of stars makes a lovely imaginary map for this story.

8

London: A City Full of Disagreement



A famous city, a change in transaction fees, and some strong opinions. Welcome to London!



Ethereum Cat Herders' London upgrade community call, August 2021, with one of the first EIP-1559 transactions on the Ropsten testnet on screen.

After **Berlin**, Ethereum's next stop was **London**, named after the city that hosted Devcon 1 in 2015. But unlike a pleasant visit to London, this upgrade came with some serious disagreements.

The London upgrade is best remembered for **EIP-1559**, a proposal to change how Ethereum transaction fees worked. Before London, users often had to guess how much gas to pay to get their transactions included in a block. EIP-1559 introduced a base fee that adjusted automatically depending on network demand. Instead of going to miners, this base fee would be burned, while miners could still receive priority fees and other rewards.

Sounds like a useful improvement, doesn't it? Well, not everyone thought so.

At the time, Ethereum was still running on Proof of Work, where miners played an important role in securing the network. Changing the fee mechanism also meant changing part of their income. Many miners opposed the proposal, while wallets, applications, and other ecosystem participants saw potential benefits in more predictable transaction fees.

EIP-1559 had been discussed for years, but technical research and implementation were only part of the challenge. Getting people to understand

the proposal, its benefits, and its possible consequences was another.

✦ When We Asked the Community

I worked with Tim Beiko through Ethereum Cat Herders on community outreach for EIP-1559. We reached out to miners, mining pools, wallets, applications, exchanges, and infrastructure providers. The idea was simple: before moving ahead with such a significant change, we wanted to understand what different stakeholders thought about it and why.

We prepared a questionnaire and arranged one-on-one conversations with several projects. In October 2020, we published the EIP-1559 Community Outreach Report, recording feedback from 25 participating projects and mining operations.

The results were eye-opening. Among the responses included in the sentiment breakdown, 42% were positive, 42% negative, and 17% neutral or undecided. The percentages are rounded, but the message was clear. Opinion was almost equally divided. Looking closer, however, told a different story. Excluding miners, 60% of respondents had a positive impression of EIP-1559. Among miners, 89% expressed a negative view.

These numbers were not an ecosystem-wide vote. They represented the mining pools and projects that participated in our outreach. Still, they helped us understand where the concerns were coming from.

For many projects, predictable gas fees were a welcome improvement. It could make life easier for wallets and applications that calculate transaction fees on behalf of users. Others appreciated the idea of burning ETH. But miners were worried about losing revenue, and some questioned whether the economic effects of EIP-1559 could be properly tested before going live.

Some projects were hesitant for another reason. They wanted to see the proposal officially included in an upgrade before spending time and resources changing their software. Others preferred not to get involved in the disagreement at all.

This taught me something important. Even when a proposal looks useful, people may have very different reasons for supporting it, opposing it, or simply waiting.

✦ Bringing the Disagreement Into the Conversation

The outreach report was not the end of the discussion. On February 26, 2021, Ethereum Cat Herders brought stakeholders with different opinions on EIP-1559 into one Zoom room. Hudson Jameson moderated the discussion while I handled the livestream. Miners shared concerns about revenue, researchers explained the economics, and developers discussed implementation and testing.

The goal was not to vote or convince everyone to agree, but to give everyone a chance to be heard. Not everyone changed their mind, of course! But the discussion helped people understand the different viewpoints and concerns. After all, open discussion may not always lead to agreement, but it is an important part of how Ethereum moves forward.

✦ More Than One EIP

As London approached, Ethereum Cat Herders hosted another community call on August 2, 2021. This time, we brought together EIP authors, client developers, researchers, wallet teams, and infrastructure providers to discuss the upcoming upgrade and their preparations.

One key point from the call was how much work could be hidden behind a seemingly simple proposal. Martin Holst Swende from the Geth team explained that EIP-1559 affected much more than transaction fees. Developers also had to update transaction handling, client behaviour, mining logic, and interfaces used by applications.

And while EIP-1559 received most of the attention, London actually included five EIPs.

I remember an earlier PEEPanEIP conversation with Vitalik Buterin and Martin Swende about EIP-3403, which proposed changing Ethereum's gas refund mechanism. At the time, gas tokens allowed people to take advantage of refunds, almost like storing gas in a battery to use later. This could create unnecessary activity and put additional pressure on the network.

Vitalik explained the importance of addressing these problems without adding too much complexity, especially with **The Merge** already being a major priority. EIP-3403 did not make it into London, but another proposal, EIP-3529, took a simpler approach to reducing gas refunds.

Alongside EIP-1559 and EIP-3529, London included EIP-3198, which allowed smart contracts to read the base fee; EIP-3541, which reserved space for future improvements to Ethereum's virtual machine; and EIP-3554, which delayed the **difficulty bomb** to give developers more time to prepare for The Merge.

During the community call, researcher Ansgar Dietrichs also encouraged people to look beyond the popular proposals. Some of the less-discussed changes could turn out to be just as important in the long run.

A good reminder that the most talked-about EIP is not necessarily the only important one. And sometimes, even a proposal that doesn't make it into an upgrade helps shape the one that does!

✦ London Finally Arrives

After years of research, discussions, testing, and preparation, London finally activated on August 5, 2021, at block 12,965,000.

Not everyone was happy with EIP-1559, even after the upgrade. But Ethereum had moved forward with a change that had been widely debated, tested, and implemented across multiple client teams.

I often think about the difference between EIP-1559 (London) and ProgPoW (EIP-1057). ProgPoW received serious technical attention and audits but never made it into an upgrade amid wider disagreement. EIP-1559 also faced strong opposition, yet eventually became part of Ethereum. Both experiences taught me something important - a proposal can be technically interesting without being adopted, and a change can move forward without everyone agreeing.

Rough Consensus Is Not a Unanimous Cheer

This is where **rough consensus** comes in. Ethereum does not decide protocol changes by simply counting votes. Discussions consider technical merits, trade-

offs, stakeholder concerns, and whether client teams are willing to implement a change. Rough consensus does not mean ignoring opposition, but understanding disagreements and finding enough common ground to move forward. Even then, the wider ecosystem must choose to adopt the upgrade.

For me, London was an important lesson in Ethereum governance. Community outreach was not just about announcing an upgrade, but listening to concerns, documenting different opinions, and helping people participate in the discussion.

After London, Ethereum Cat Herders recognized contributors with commemorative NFTs. I remember sharing a thought during our community celebrations - “I believe that a blockchain is decentralized in true sense when every milestone is celebrated with the community.”

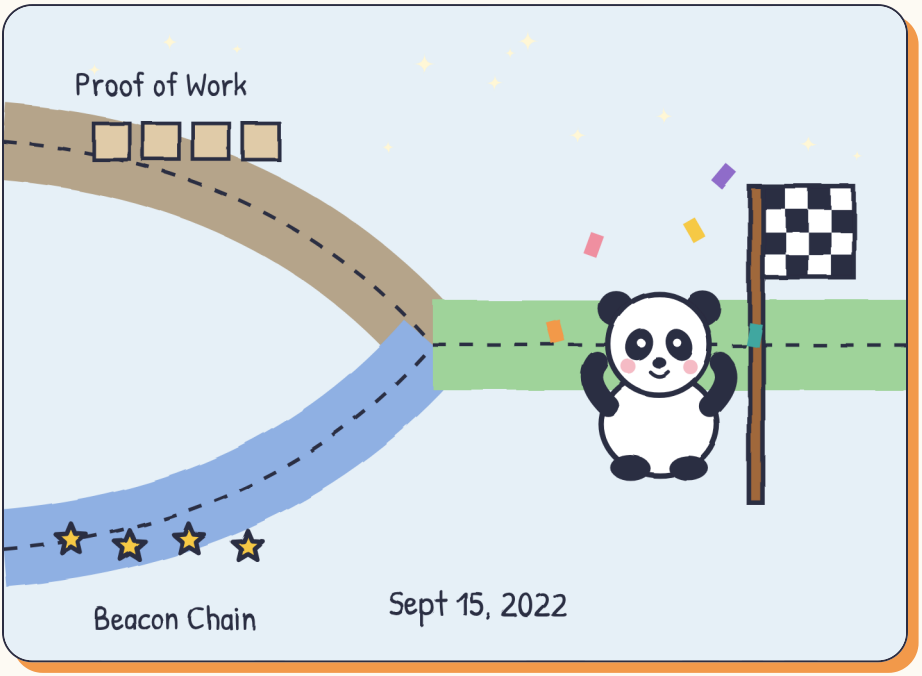
London took a lot of people, and they did not all agree on everything. Perhaps that’s the beauty of Ethereum. Consensus doesn’t always mean everyone agrees, but finding a way forward despite disagreements.

FUN FACT ✦

London was named after the city that hosted Devcon 1, but its most famous proposal was about transaction fees, not travel! And while EIP-1559 took the spotlight, four other EIPs travelled with it. One city, five proposals, and plenty of conversations along the way.

9

Two Worlds, One Ethereum



This time, two worlds come together. And there's a panda waiting at the finish line!



"Activated!" The official Mainnet Merge livestream, September 15, 2022, which I co-hosted. Watch it at youtube.com/watch?v=Nx-jYgI0QVI



"PoS Activated," framed by two pandas, while the community celebrates on the same livestream.

By September 2022, Ethereum had spent years preparing for one of its biggest changes - moving from Proof of Work to Proof of Stake. The **Beacon Chain** was

already running, developers had tested the transition many times, and everyone wanted to make sure Ethereum could make the switch safely.

The transition happened in two stages. First came **Bellatrix**, the Consensus Layer upgrade, on September 6. Then came **Paris**, the Execution Layer transition, on September 15. Together, they formed what we know as **The Merge**.

Unlike later upgrades, Bellatrix and Paris were not combined into a new name. There was no Bellaris or Paritrix. It was simply called “The Merge”. And that name said it all!

For me, that night is hard to forget. It was around 2:42 in the morning on the East Coast of the United States. I was one of the livestream hosts, talking with developers, researchers, and guests about what was happening and what might come next.

And then, the panda appeared on the screen!

Ethereum had successfully moved from Proof of Work to Proof of Stake. After years of preparation, the moment everyone had been waiting for had finally arrived.

What surprised me was how little changed for someone simply using Ethereum. There was no “blockchain under upgrade” or dramatic pause or a new Ethereum starting from scratch. Blocks and transactions continued as the network switched its consensus mechanism. Developers sometimes joke that the best network upgrade is a boring one. And in this case, boring was a very good thing!

Another important part of preparing for The Merge was client diversity. Ethereum relies on different software clients following the same rules. If too many people use the same client, a serious bug could put a large part of the network at risk. Encouraging different clients was an important part of making the transition safer.

When I watch that livestream today, it almost feels like a different lifetime. But I still remember the excitement, the long preparation, and the relief of watching Ethereum continue without interruption.

The panda was a mascot, not a protocol rule!

And what about our panda? Well, the software certainly didn't need one to complete The Merge. But for the community watching history unfold, a cute panda was a wonderful way to celebrate!



From PeepaneIP ✦

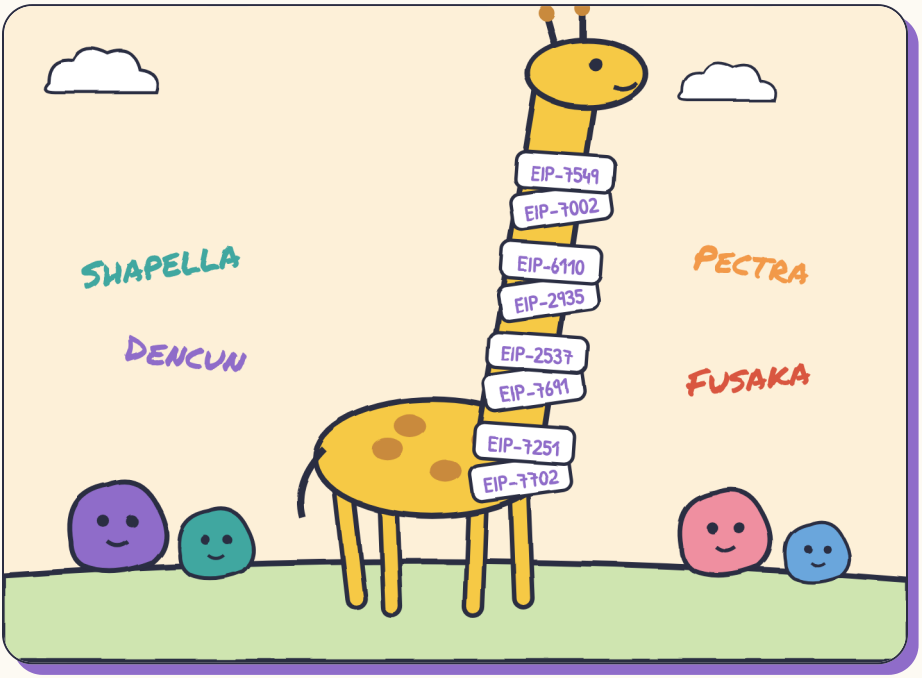
When I later asked Vitalik Buterin which EIP or change excited him most, his answer was The Merge. After years of work, I can understand why that was the one he chose.

Fun Fact ✦

The panda became the mascot of The Merge. It brought a little fun to one of Ethereum's biggest technical achievements. Sometimes, even years of serious engineering deserve a cute celebration!

10

Shapella, Blobs and a Very Tall Giraffe



Hello, Etherean! ✦

> We have met cities and stars separately. But what happens when they come together?



The official Dencun mainnet watch party, March 13, 2024, which I co-hosted with EthStaker and The Daily Gwei. Watch it at youtube.com/watch?v=iL0cZRkyrV0



The official Pectra mainnet upgrade watch party, May 7, 2025, which I co-hosted. Watch it at x.com/i/broadcasts/1kvJpyZdeLbxE

After **The Merge**, Ethereum had two naming traditions to follow. Execution Layer upgrades were named after Devcon cities, while Consensus Layer

upgrades followed star names. Since both layers now worked together, combining their names became a useful way to talk about an upgrade affecting the entire network.

The first such combination was **Shapella**, which went live on April 12, 2023. Shanghai, named after the city that hosted Devcon 2, joined **Capella**, the brightest star in the constellation Auriga.

I remember explaining this to a journalist who was asking about the Shanghai upgrade. I mentioned Shapella, and the next question was obvious: "What is Shapella?" A fair question! Shanghai was only one part of the upgrade. Capella was the other.

Shapella was an important milestone after The Merge. It enabled validators to withdraw their staked ETH, something they had been waiting for since Ethereum introduced staking. EIP-4895 helped make those withdrawals possible.

Ethereum Cat Herders and EthStaker hosted a viewing party to celebrate the upgrade. These watch parties had become a tradition of their own, with developers, validators, and community members coming together to watch Ethereum reach another milestone. Almost like waiting for a concert to begin, except the main performer was a blockchain upgrade!

✦ Next Stop: Dencun and the Blobs

In March 2024, another city met another star. Deneb, a bright star in the constellation Cygnus, joined Cancun, the host city of Devcon 3. Together, they became **Dencun**.

This upgrade brought something new to Ethereum: **blobs**!

No, not jelly or a blob monster. Blobs are temporary data storage introduced through **EIP-4844** to help Layer 2 networks, or rollups, publish data to Ethereum at a lower cost. Before Dencun, posting this data could be expensive. Blobs provided a cheaper way to do it, helping reduce transaction costs for many Layer 2 users.

After nearly two years of discussions, testing, and development, Dencun went live on March 13, 2024. It was another major step towards making Ethereum

more scalable and affordable.

✦ Pectra and the Very Tall Giraffe

Next came **Pectra** in May 2025. This time, Prague, the city that hosted Devcon IV, joined Electra, a giant star in the constellation Taurus. Pectra brought several improvements, including **EIP-7702** for smarter Ethereum accounts, EIP-7251 for validator balances, and EIP-7691 to increase blob capacity.

But the upgrade had something else that caught the community's attention: a giraffe!

As discussions continued and the list of candidate EIPs grew, the giraffe's neck became a playful way to represent the growing scope of the upgrade. The longer the list, the taller the giraffe neck!

Of course, not every proposed EIP made it into Pectra. But the giraffe gave the community something fun to remember while developers worked through the technical decisions.

✦ Fusaka: More Room for Blobs

By December 2025, Ethereum was ready for another upgrade. Fulu, a star in the constellation Cassiopeia, joined Osaka, the city that hosted Devcon V. Together, they became **Fusaka**.

Fusaka continued the scaling journey that began with Dencun. Its main feature, **PeerDAS** (EIP-7594), introduced a way for Ethereum nodes to check whether blob data was available by sampling pieces of it instead of downloading everything.

Think of it like checking a few pages of a very large book rather than carrying the entire book around. The technology is more complicated, of course, but the goal was simple: allow Ethereum to handle more blob data without asking every node to download all of it.

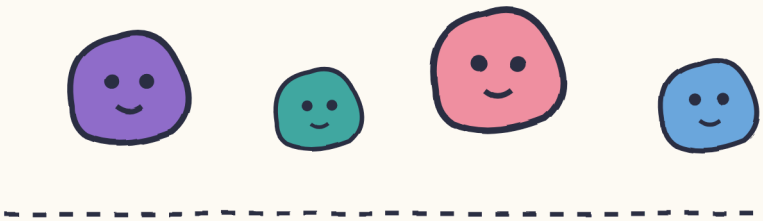
Fusaka went live on December 3, 2025. It also introduced a different approach to increasing blob capacity through smaller, focused upgrades called Blob-Parameter-Only upgrades.

THE RECORD ✦ Ethereum's Portmanteau Names

Upgrade	Execution Layer (city)	Consensus Layer (star)	Mainnet
Shapella	Shanghai (Devcon 2)	Capella (Auriga)	April 12, 2023
Dencun	Cancun (Devcon 3)	Deneb (Cygnus)	March 13, 2024
Pectra	Prague (Devcon IV)	Electra (Taurus)	May 7, 2025
Fusaka	Osaka (Devcon V)	Fulu (Cassiopeia)	December 3, 2025

Four upgrades, four cities, and four stars. What started as two separate naming traditions had now become a familiar pattern. And as Ethereum continued to evolve, the community found new ways to name and remember its upgrades. But not every upgrade needed a city or a star. Some were happy with just a number. And one star even had to change its name! More on that in our next chapter.

And yes, Ethereum upgrades have had their share of animal mascots too. Who knew protocol upgrades could be this entertaining?



FUN FACT ✦

A blob is not jelly! It is temporary data space that helps rollups publish information more efficiently.

11

The Missing Star and the Numbered Upgrades



Even stars need a name check. And some upgrades are perfectly happy with numbers!

By the time Ethereum reached the letter H in its Consensus Layer star-naming sequence, upgrade names had become something people paid attention to.

The initial choice was **Heka**. But someone raised an interesting question. Did Heka actually meet the star-naming convention being discussed for Ethereum upgrades?

As the community looked into it, they found that Heka did not meet the proposed requirement for an officially recognized star name. The name was reconsidered, and **Heze** was selected for the Consensus Layer upgrade paired with Bogotá. Together, they became **Hegotá**.

A good example of how even an upgrade name can change when someone checks the details. No cosmic disaster, just a correction!

But does every Ethereum upgrade need a creative name? Not really. Sometimes, keeping it simple works better.

As Ethereum continued working on blob scaling, a different kind of upgrade appeared: Blob-Parameter-Only (BPO) upgrades. The idea, described in EIP-7892, was to allow changes to selected blob parameters without waiting for a larger network upgrade. These focused changes could help Ethereum adjust blob capacity as needed.

These upgrades were simply called **BPO1**, BPO2, and so on. No glacier, no star, no city name. Just a number.

It might sound less exciting, but not every upgrade needs a fancy name. When changes are focused and repeatable, simple numbering makes sense.

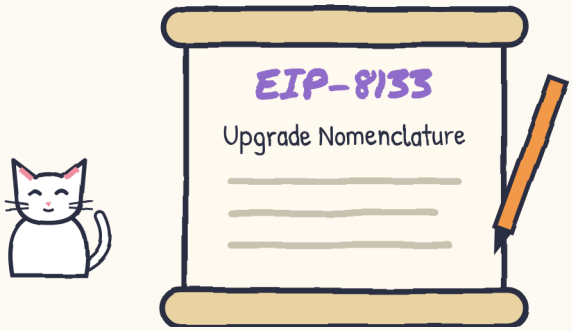
While documenting BPO upgrades, I started looking more closely at all the naming patterns Ethereum had collected over the years. Why cities? Why stars? Why glaciers? Why combine names? And more importantly, where could someone new to Ethereum find the story behind them?

That curiosity led me to write **EIP-8133**, an Informational EIP documenting Ethereum's upgrade naming conventions.

Its purpose is simple: to provide a reference by documenting the traditions, how they developed, and how they are being used. Future names are still chosen by

the community, in open discussions that anyone can join. So the next name could come from you! After all, documenting a community practice helps everyone take part in a decentralized ecosystem.

That's one of the lovely things about Ethereum's story. Even something as small as an upgrade name can lead to a discussion, a correction, or a new EIP.



FUN FACT ✦

BPO stands for Blob-Parameter-Only. Sometimes the most useful upgrade name is simply a number, BPO1, BPO2...

12

What Do We Call Ethereum's Next Upgrade?



Our passport is full. But Ethereum's story is far from finished.

Ethereum has travelled from a frontier to a homestead, met a whistle and a dragon, visited cities, crossed glaciers, counted stars, and watched a panda celebrate. Along the way, something even more important happened. Ethereum found its people, and together they built an ecosystem. A community that proposes ideas, developers who test code, stakeholders who raise objections, and contributors who explain difficult changes. And yes, people who sometimes change their minds.

Not every proposal makes it into an upgrade. ProgPoW did not. **EIP-1559** did, despite serious disagreements. Account abstraction found a path through ERC-4337 before **EIP-7702** reached **Pectra**. **The Merge** took years of preparation. These stories remind us that Ethereum's evolution is not always straightforward.

Upgrade names are just one small part of this larger story. They become memorable because each change adds something to Ethereum's shared history. I believe Ethereum will become what its ecosystem wants it to be. The community continues to shape it through EIPs, discussions, and contributions.

Of course, that doesn't mean anyone can change the protocol simply by wishing for it. But anyone can ask questions, learn the process, share ideas, contribute evidence, and help others understand what is happening.

So, the next time you hear a strange Ethereum upgrade name, ask where it came from. Who knows? The answer might lead you to an EIP, a developer meeting, a community discussion, or perhaps an entirely new question.

FROM VITALIK'S CONVERSATION ✦

During one of my PEEPanEIP conversations with Vitalik Buterin, I asked him a few quick questions about Ethereum's future. Where did he see Ethereum in 2030? He imagined a network with more scale, more applications, better privacy, and a much better user experience. A global platform for building an open, decentralized world.

I also asked what Ethereum should never compromise on. His answer was simple: "Openness, decentralization, censorship resistance." These are

important reminders that, beyond all the upgrades and funny names, Ethereum is about much more than technology.

Of course, I couldn't resist asking him a playful question. If he could rename Ethereum, what would he call it?

He wouldn't change the name Ethereum, but he liked the idea of reviving an old proposal (EIP-6789) to rename gas. Perhaps we'll have a different name for gas someday. Gas or Mana, what would you choose?

And when I asked what Ethereum's spirit animal would be, his answer was "Phoenix". Quite fitting for a network that keeps evolving, isn't it?

Before we close our passport, here's one last thought. The next great Ethereum upgrade might begin with someone asking, "What if?" Not every idea becomes an EIP, and not every EIP becomes part of an upgrade. But every idea has the potential to start a conversation.

And perhaps one day, when you notice something Ethereum could do better, you will ask your own "What if?" That question could be the beginning of another EIP, another conversation, or a future upgrade whose name we have not heard yet. You do not need to be a core developer to begin asking.

✦ Ethertastic, isn't it? ✦

We started this journey with a simple hello. Now that you've travelled through Ethereum's cities, stars, glaciers, and dragons, perhaps the next chapter will have a little of your story in it too.

Your Turn: Make a Name

Hello again, Etherean. Want to make-believe with me? Imagine your own little constellation of upgrade names. What would you call the next stop?

Choose a city from a real Ethereum event and a star name that fits the current naming convention. This is a creative exercise, not an official proposal.

My city:

My star:

My combined name:

Why I chose it:





your star here?





Timeline of Ethereum Upgrades

- 2015 **Frontier** Jul 30
  Ethereum's first live network
- 2015 **Frontier Thawing** Sep 7
  Gas limit restriction lifted
- 2016 **Homestead** Mar 14
  Pi Day, block 1,150,000
- 2016 **DAO Fork** Jul 20
  Ethereum and Ethereum Classic part ways
- 2016 **Tangerine Whistle** Oct 18
  Anti-DoS emergency upgrade
- 2016 **Spurious Dragon** Nov 22
  State clearing and replay protection
- 2017 **Byzantium** Oct 16
  First part of Metropolis
- 2019 **Constantinople + Petersburg** Feb 28
  Two upgrades, one block
- 2019 **Istanbul** Dec 7
  Last of the historical city trio
- 2020 **Muir Glacier** Jan
  First glacier, difficulty bomb delayed
- 2020 **Beacon Chain** Dec 1
  Proof of Stake foundation launches
- 2021 **Berlin** Apr 15
  First Devcon city name
- 2021 **London** Aug 5
  EIP-1559 and four more EIPs
- 2021 **Altair** Oct 27
  First star, first Beacon Chain upgrade
- 2021 **Arrow Glacier** Dec
  Another bomb delay

2022 **Gray Glacier** Jun
A glacier that merges

2022 **The Merge (Bellatrix + Paris)** Sep 6 / 15
Proof of Stake, and a panda

2023 **Shapella** Apr 12
Validator withdrawals

2024 **Dencun** Mar 13
Blobs arrive

2025 **Pectra** May 7
Smarter accounts, a very tall giraffe

2025 **Fusaka** Dec 3
PeerDAS and BPO upgrades

2026 **Glamsterdam** Sepolia ✓
Mainnet date not yet confirmed (Oct 2026)

NEXT **Hegotá** In development
Heze + Bogotá

Glossary

Beacon Chain
The Proof-of-Stake chain launched in December 2020 that became Ethereum's Consensus Layer at The Merge.

Blob
Temporary data space added in Dencun (EIP-4844) that helps rollups publish data more cheaply.

BPO upgrade
Blob-Parameter-Only upgrade (EIP-7892). A small upgrade that only changes blob settings, named by number: BPO1, BPO2...

Client
Software that runs Ethereum. Different teams build different clients that follow the same rules.

Consensus Layer
The part of Ethereum that agrees on the chain. Its upgrades are named after stars, in alphabetical order.

Difficulty bomb
A Proof-of-Work mechanism that slowly made mining harder, creating the "Ice Age."

EIP
Ethereum Improvement Proposal. A document describing a proposed change or a standard.

Execution Layer
The part of Ethereum that runs transactions and smart contracts. Its upgrades are named after Devcon and Devconnect cities.

Gas
The unit that measures the computing work a transaction needs.

Hard fork
A change to Ethereum's rules that every node must adopt to stay on the same chain.

Meta EIP
An EIP that lists what is included in a network upgrade.

Portmanteau
A word made by combining parts of two words, like Shapella or Glamsterdam.

Proof of Stake / Proof of Work
Two ways of securing a blockchain: validators staking ETH, or miners spending computing power.

Rollup
A Layer 2 network that processes transactions away from Ethereum and posts data back to it.

Rough consensus
Moving forward when concerns are understood and enough common ground exists, without needing a unanimous vote.

Validator
A participant who stakes ETH to help propose and confirm blocks.

Related EIPs

Read any of these at eips.ethereum.org.

EIP-2, 7, 8
Homestead changes

EIP-150
Gas cost changes (Tangerine Whistle)

EIP-155
Replay protection

EIP-161
State clearing

EIP-607
Meta: Spurious Dragon

EIP-608
Meta: Tangerine Whistle

EIP-779
DAO Fork

EIP-1013
Meta: Constantinople

EIP-1057
ProgPoW (not adopted)

EIP-1283
Net gas metering (removed by Petersburg)

EIP-1559
Fee market change

EIP-1679
Meta: Istanbul

EIP-1716
Meta: Petersburg

EIP-2384
Muir Glacier bomb delay

EIP-2565
ModExp gas cost

EIP-2718
Typed transactions

EIP-2929
State access gas costs

EIP-2930
Access lists

EIP-3198
BASEFEE opcode

EIP-3403
Partial refund removal (not included)

EIP-3529
Reduced gas refunds

EIP-3541
Reserve 0xEF prefix

EIP-3554
Bomb delay (London)

ERC-4337
Account abstraction

EIP-4345
Arrow Glacier bomb delay

EIP-4844
Blobs (proto-danksharding)

EIP-4895
Validator withdrawals

EIP-5133
Gray Glacier bomb delay

EIP-6789
Rename gas to mana

EIP-7251
Larger validator balances

EIP-7594
PeerDAS

EIP-7691
Blob throughput increase

EIP-7702
Smarter accounts

EIP-7723
Network upgrade inclusion process

EIP-7892
Blob-Parameter-Only upgrades

EIP-8133
Upgrade nomenclature

Sources

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- Ethereum Cat Herders, Constantinople postponement post-mortem, January 2019
- All Core Devs meeting #68 notes, August 2019, github.com/ethereum/pm
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- Ethereum Foundation Blog: Gray Glacier announcement, June 2022
- Public GitHub discussion on the Bellatrix name
- PEEPanEIP episodes with Vitalik Buterin, Martin Holst Swende, Micah Zoltu, Danny Ryan, Alex Stokes, James Hancock and Thomas Jay Rush
- The Code Confession, conversation with Thomas Jay Rush
- Mainnet Merge livestream, September 15, 2022: youtube.com/watch?v=Nx-jYgI0QVI
- Dencun mainnet watch party, March 13, 2024: youtube.com/watch?v=iL0cZRkyrV0
- Pectra mainnet watch party, May 7, 2025: x.com/i/broadcasts/1kvJpyZdeLbxE
- My own recollections from coordinating upgrades with Ethereum Cat Herders

Thank You

This book would not exist without the people who build, question, and celebrate Ethereum.

Thank you to every guest who joined me on PEEPanEIP and The Code Confession and generously shared their time, knowledge, and memories, including Vitalik Buterin, Martin Holst Swende, Micah Zoltu, Danny Ryan, Alex Stokes, James Hancock, Thomas Jay Rush, Ben Edgington, and many more.

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And thank you, reader, for being curious enough to ask where a name comes from.

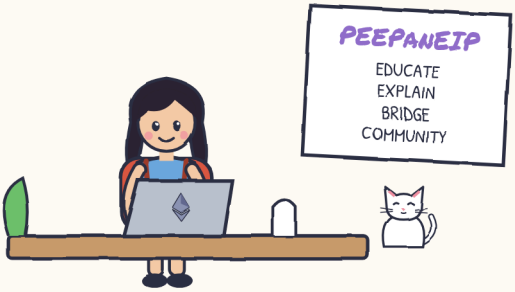
Feedback is a gift ✦

Did you enjoy this book? Did a story remind you of something you saw happen? Your feedback is the greatest gift. Tag me on X to share what you liked, what could be better, or a historical anecdote of your own. The best stories may find their way into the next edition!

@poojaranjan19

About the Author





Pooja Ranjan is an Ethereum community member, researcher, and educator who has been part of the ecosystem since 2016.

She is the founder of Avarch, President of ECH Institute (formerly Ethereum Cat Herders), and the EIP Coordinator, helping bring people, proposals, and processes together to support Ethereum's evolution.

She co-authored EIP-5069 (the EIP Editor Handbook), authored EIP-8133 on Ethereum upgrade naming conventions, and has contributed to several other EIPs.

Pooja is also the creator of **PEEPanEIP**, an educational video series, and host of **The Code Confession**, where she explores the people and stories behind Ethereum.

She believes that asking questions, connecting people, and making complex ideas simple are all part of keeping Ethereum open and welcoming to everyone.

Draw a Mascot

The Merge had a panda. Pectra had a very tall giraffe. What mascot would you give the next upgrade? Draw it here and give it a name!

My mascot's name:

NOTES

More Notes

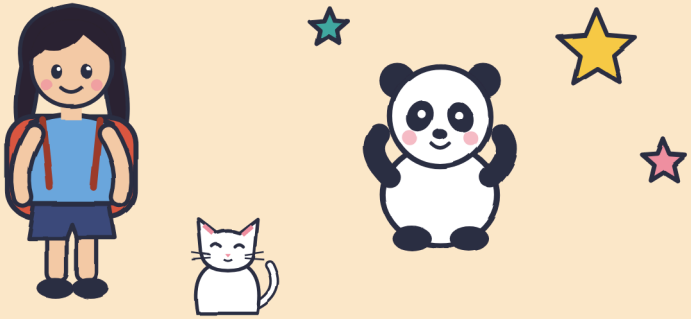
Signatures + Stamps

Met an EIP author, a client developer or a fellow Etherean? Ask them to sign here.



Why is an Ethereum upgrade called **Glamsterdam**?
Who named the **Spurious Dragon**?
And why did a star have to change its name?

Travel from the Frontier to the stars in twelve short, illustrated stories about the people, proposals and peculiar names behind Ethereum's upgrades, told by someone who helped coordinate many of them.



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